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# Improving Health Information Systems for Universal Health Coverage in the Philippines

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## ABSTRACT

Health information systems are a key building block of a well-functioning health system and are a vital component in the implementation of universal health coverage. The Philippines's current transition toward universal health coverage had attendant health information system reforms, such as the establishment of the country's National Health Data Repository and the Health Sector Enterprise Architecture. However, many issues remain in this transition, such as insufficient investments in health information systems, poor reliability of health information systems, technical support delays, data quality issues, and cybersecurity issues. We recommend making health data portable throughout the health system while adhering to data privacy laws, investing in internet connectivity, information technology, and capacity building for human resources, and improving implementation of current policies and guidelines to fully harness the potential of health information systems in making universal health coverage a reality in the Philippines.

## 1 | Introduction

A well-functioning health information system (HIS) is a key building block of a strong health system. It is essential in keeping track of patient history, in disease surveillance, in processing of claims and co-payments, in predicting and forecasting demand for services, and in the monitoring, evaluation, and continued improvements of health policies and programs [1]. Without an integrated and responsive HIS, it is difficult to monitor patient pathways for continuity of care, conduct disease surveillance, process provider payments, forecast demand for services and resources, and track health systems performance [1].

Despite substantial advancements in HIS and the ubiquity of electronic medical records (EMRs), data availability for universal health coverage (UHC) indicators is only 70% in the Western Pacific Region, which affects healthcare planning and service delivery. For this reason, the Global Strategy on Digital Health 2020–2025 recognizes that a responsive HIS is a catalyst to meeting UHC and global health targets, whereas acknowledging the challenges faced by low- and middle-income countries (LMICs) with significant resource, governance, health system, behavioral, organizational, and technical constraints [2].

This paper aims to document recent HIS reforms in the Philippines, describe challenges in health information management

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in the country, and put forth some recommendations to address these challenges as the country implements reforms toward UHC. In doing so, we hope to push for reform in health information management in the Philippines and in other countries encountering health information management challenges as they transition toward UHC.

## 2 | UHC and Recent HIS Reforms in the Philippines

In the Philippines, the passage of the UHC Law in 2019 ushered in rapid developments in the country's HIS environment. As mandated in its implementing rules and regulations, all health service providers and insurers are now required to maintain an HIS covering enterprise resource, human resources for health, EMRs, and electronic prescription log, which shall be electronically uploaded on a regular basis through an interoperable system called the National Health Data Repository (NHDR) [3]. The NHDR is envisioned as the authoritative repository of the country's health and health-related data. It will serve as the basis of health policy and standards, decision-making, health program planning and implementation, health systems strengthening, and monitoring and evaluation. Furthermore, the Health Sector Enterprise Architecture (HSEA) is being developed to standardize and organize the information and communications technology infrastructure of the country, ensure its alignment to the aim of UHC, and support digital transformation of the healthcare sector. These reforms are key to unify and harmonize HIS in the country, with standards enforced to ensure quality and interoperability across various public and private systems [4].

## 3 | Health Information Management Challenges in the Philippines

### 3.1 | Insufficient Investments in HISs

In a rapid survey we conducted with 20 purposively-selected local government units (LGUs) in the Philippines for project planning purposes (Supporting Information File 1), only 10 (50%) have budget allotment for HIS operations and maintenance, and only one mentioned having dedicated space, IT personnel, and a permanent trained encoder running the LGU's HIS. Data management and encoding of health information from many government facilities are still being done by midwives, nurses, and physicians, increasing their workload and giving them less time to care for patients. In many less-urbanized areas in the country, paper-based recording and reporting of health events are still prevalent. The lack of HIS investments of LGUs highlights the unrecognized importance of enhancing capacities and infrastructure for health data management. Because of the shortages in human and information technology resources due to chronic underinvestment, many LGUs will have difficulty implementing these new HIS reforms.

### 3.2 | Poor Reliability of HISs

Many of our survey respondents reported that the HIS they are using commonly experience server downtimes, which is due to poor internet connectivity or bugs in the system. Some HIS platforms are still under development and require systems updates, which would necessitate retraining for users. Lastly, respondents also reported the HIS platform's inability to operate as a standalone system as a challenge, indicating the need to have various platforms installed to generate different reports or process claims.

### 3.3 | Technical Support Delays

In the same survey, only three (15%) have dedicated local technical support teams; others have to call a central office technical support team, which could mean significant delays. Seven (35%) LGUs receive response to their technical support queries within the day, five (25%) receive response within 1 to 2 days, whereas six (30%) receive technical support anywhere from 3 to 5 days. One (5%) LGU, located in a mountainous area in the Philippines, reported that they waited for more than a month before receiving technical support. These delays create inefficiencies in the healthcare sector, with health providers reverting to paper-based data collection and management.

### 3.4 | Data Quality Issues

Of 20 respondent LGUs, 12 (60%) do not have an established protocol for validating the data collected, raising concerns about data quality. Among the eight (40%) LGUs that did, data validation methods included reviewing data during input, interviewing patients, or during program implementation and/or system review. These quality issues affect estimates on disease burden and UHC indicators such as coverage and quality of care, which ultimately impact how services are designed, promoted, delivered, and financed [5, 6].

### 3.5 | Lack of Data Portability and Interoperability of HIS

Ideally, when a patient is referred from one hospital to another, his/her data should accompany him/her as well, providing continuity of care from one health facility to another. However, health data ownership in the Philippines typically lies with the providers, and patients will have to get permission from the provider to share their health data when they go to another facility [7]. Further aggravating this issue is the lack of interoperability of the different HIS currently existing in the country, despite the guidelines explicitly stating that they have to be interoperable [8]. As a result, the patient may not have continuity of care as he/she navigates through the health system.

### 3.6 | Cybersecurity Issues

There have been many verified cybersecurity breaches of health and insurance data in the Philippines [9]. Beyond these national agencies, we have also identified cybersecurity vulnerabilities in our respondent LGUs. In our survey, 12 (60%) LGUs store health data in only one platform—either their desktop computer in the health facility or a server linked to the EMR. Implementing



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storage and data redundancy is essential in securing sensitive individual health data and preventing possible data loss.

## 4 | Recommendations

The promise of UHC in the Philippines through HIS reforms is tremendous; however, many challenges remain that require a whole-of-society approach to address [2, 10]. We provide recommendations backed by successful experience from other countries:

### 4.1 | Making Health Data Portable Throughout the Health System While Adhering to Data Privacy Laws

Previous research studies have highlighted the vital role of standardized data collection and transmission systems, universal identifiers for individual patients, and modernized working practices to facilitate health data portability as patients navigate throughout the health system [11]. In the Philippines, the closest universal identifier that could be used is the national identification system, and it can be integrated into the country's various HISs to facilitate UHC implementation; however, data privacy concerns persist, which preclude its use. The approach used by some HISs in Europe, in compliance with their General Data Protection Regulation, includes *k*-anonymization, masking patient and admission identifiers, deleting all direct identifiers, and removing images and free text except for laboratory variables [12].

### 4.2 | Investing in Internet Connectivity, IT, and Capacity Building for the Health Workforce

A key facilitator during the early phase of transition to electronic HIS in the United States was substantial financial investments, which allocated federal budget for IT resources and training their health workforce with the overall goal of improving quality of care [13]. However, in the Philippine context, financial investments should come not just from the national government but also from the local government, which is empowered through the Local Government Code to design and manage their own health systems [14] and, as a result, their own local HIS. Through guidance from the Department of Interior and Local Government, there could be roadmaps for LGUs for transitioning from a paper-based system toward electronic HIS. These roadmaps should describe which IT infrastructure and training needs should be addressed at each stage of the transition [2]. This will also entail concomitant investments from private telecommunication providers, to ensure reliable internet connectivity even in less urbanized areas. To encourage the private sector to do this, tax breaks or government subsidies could be given.

### 4.3 | Improving Implementation of Current Policies

Guidelines for interoperability, data quality, and reducing cybersecurity vulnerabilities are already present in various policy documents in the Philippines [8, 4]. However, these are not implemented properly because personal information controllers and processors in the country may not be aware of such guidelines or do not have the necessary skills or resources to implement it. This necessitates national government agencies to streamline the dissemination of pertinent guidelines for interoperability, data quality, and cybersecurity toward local government health centers and continuously invest in capacity building to implement the said guidelines.

## 5 | Conclusions

These nascent HIS reforms and the challenges in implementing them are part of the birthing pains in the transition toward an integrated health system necessary in making UHC possible. These issues and the gravity of the necessary reforms highlight the need for different national government agencies, local governments, as well as partners in the private sector to come together and invest in information technology and the infrastructure and human resources necessary to improve HIS for UHC.

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#### Author Contributions

**Catherine Agas Chung**: conceptualization, supervision, data curation, formal analysis, writing – original draft. **Jenilyn Ann Vergara Dabu**: conceptualization, data curation, formal analysis, writing – original draft. **Fiona Rose Andres Norada**: conceptualization, data curation, formal analysis, writing – original draft. **Arianna Maever Loreche**: conceptualization, formal analysis, writing – review and editing. **Veincent Christian F. Pepito**: conceptualization, formal analysis, writing – review and editing. **Manuel M. Dayrit**: conceptualization, supervision, formal analysis, writing—review and editing.

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#### Ethics Statement

The authors have nothing to report.

#### Consent

The authors have nothing to report.

#### Conflicts of Interest

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#### Data Availability Statement

The survey data mentioned in the study was conducted for program planning purposes and not for research. It may be accessed from Dr Catherine Chung (cachung@zuelligfoundation.org) upon reasonable request.



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## Supporting Information

Additional supporting information can be found online in the Supporting Information section.




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